

0330

8. Factor = Type of fertilizer
treatment / levels: The three types of fertilizers

9. Factor = Daily dose of Vitamin C
treatment / levels: 200, 500, 1000 mg

14. 每個不良品比例 $\hat{p}$ 其實是「樣本平均數」

Bernoulli $\rightarrow \hat{p} = \frac{x_1 + x_2 + \dots + x_{100}}{100}, x_i = 0 \text{ or } 1$

$n=100$. 根據 CLT, 無論像本 Bernoulli 分佈的 p 為何
只要 n 夠大, $\hat{p}$ 抽樣分佈近似常態.

25 天獨立抽樣 $\rightarrow$ 25 個 $\hat{p}_i$ 來自同個常態母體 $\Rightarrow$ 隨意假設.

8.a. $\bar{X}_A = 76$

$\bar{X}_B = 66.33$

$\bar{X}_C = 81$

總和 $X_{..} = 840$

$\sum X_{ij}^2 = 65286$

$CM = \frac{X_{..}^2}{n} = \frac{840^2}{11} = 64145.45$

$TSS = 65286 - 64145.45 = 1140.55$

$SS_T = \left(\frac{380^2}{5} + \frac{199^2}{3} + \frac{261^2}{3} \right) - 64145.45 = 64787.33 - 64145.45 = 641.88$

$SS_E = 1140.55 - 641.88 = 498.67$

ANOVA

Source	SS	df	MS	F
Treatment	641.88	2	320.94	5.15
Error	498.67	8	62.33	
Total	1140.55	10		

b. $H_0: \mu_A = \mu_B = \mu_C$, H_a : 至少有 2 組的平均分數不同

$$F_{2,8,0.05} = 4.46$$

$\because 5.15 > 4.46 \therefore \text{Reject } H_0$

The data provide sufficient evidence to indicate a difference in mean student scores

10.a. Yes, 有顯著差異.

$\because p\text{-value} = 0.0251 < 0.05 \therefore \text{Reject } H_0 (\mu_A = \mu_B = \mu_C)$

b. 99% C.I.

$$S_p^2 = \frac{3 \times 9.67 + 2 \times 9.33}{4+3-2} = 9.53$$

$$60.50 - 54.67 \pm t_{5,0.05} \sqrt{9.53 \left(\frac{1}{4} + \frac{1}{3} \right)}$$
$$= [-3.68, 15.34] \rightarrow \text{包含 } 0$$

在 $\alpha=1\%$ 下, A, B 平均沒有顯著差異.

c. 99% C.I.

$$60.5 \pm t_{3,0.005} \times \sqrt{\frac{9.67}{4}} = (51.45, 69.55)$$

d. The data satisfy the assumption that they have been selected from normal population

因其樣本數很小且無明顯的極端值, 因此 ANOVA 的常態假設大致成立